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Original Research Article

Comparative assessment of pulmonary function in non-smokers, tobacco smokers, and exclusive e-cigarette users among young adults: A cross-sectional spirometric analysis

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Abstract

Background: Both tobacco smoking and electronic nicotine delivery systems (ENDS) are associated with airway injury, yet direct three-group spirometric comparisons in young adult cohorts remain limited. This study aimed to compare pulmonary function parameters, spirometric pattern distribution, and respiratory symptom burden across non-smokers, tobacco smokers, and exclusive e-cigarette users.

Materials and Methods: An analytical cross-sectional study was conducted in the Department of Physiology. A total of 300 young adults were equally allocated into three groups: non-smokers (n = 100), tobacco smokers (n = 100), and exclusive e-cigarette users (n = 100). Spirometry was performed per ATS/ERS 2019 standardisation criteria, recording FVC, FEV_1, FEV_1/FVC, PEFR, and $FEF_{25–75}$. Between-group comparisons employed one-way ANOVA with Bonferroni post-hoc correction and Pearson chi-square with Cramér's V effect size estimation. For e-cigarette users, duration and frequency of use were recorded, whereas device wattage, nicotine formulation/concentration, and flavour-specific constituent data were not systematically available.

Results: A statistically significant three-tier gradient was observed for FEV_1/FVC (non-smokers: 83.78 ± 3.48%; e-cigarette users: 79.57 ± 5.70%; tobacco smokers: 74.22 ± 7.86%; F = 64.852; $p < 0.001$; $\eta^2 = 0.304$) and $FEF_{25–75}$ ($\eta^2 = 0.206$), with large effect sizes across all pairwise contrasts. Obstructive spirometric pattern prevalence followed the same gradient: tobacco smokers 43%, e-cigarette users 24%, and non-smokers 4% ($\chi^2 = 59.40$; $p < 0.001$; Cramér's V = 0.315). FVC and PEFR did not differ significantly across groups. Respiratory symptoms showed no statistically significant between-group differences despite clear spirometric impairment.

Conclusion: Both tobacco smoking and e-cigarette use produced measurable, dose-ordered spirometric impairment in young adults, with a consistent non-smoker > e-cigarette > tobacco smoker gradient. The selective reduction in $FEF_{25–75}$ in both exposed groups is compatible with early peripheral small-airway dysfunction preceding symptom differentiation, underscoring the potential value of routine spirometric screening in young nicotine-exposed individuals.

Keywords: Spirometry; Electronic cigarettes; Tobacco smoking; Pulmonary function tests; Airflow obstruction

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1. Introduction

Tobacco smoking remains among the most significant preventable causes of chronic respiratory morbidity worldwide, with over 1.3 billion current users globally according to the World Health Organisation.[1] The lungs bear the primary burden of combustible smoke exposure — a complex mixture of thousands of chemicals, oxidants, particulates, and carcinogens — which drives progressive oxidative stress, epithelial injury, chronic airway inflammation, and structural small-airway remodelling, ultimately manifesting as measurable airflow limitation on spirometry.[2] Even in adults without established lung disease, longitudinal cohort data show accelerated FEV_1 decline in former and current smokers, including those with relatively low cumulative smoking exposure.[3]

Alongside this established tobacco epidemic, electronic nicotine delivery systems (ENDS) have expanded rapidly among adolescents and young adults worldwide.[4] Unlike combustible cigarettes, e-cigarettes generate aerosols by heating propylene glycol, vegetable glycerin, nicotine, and flavouring agents — a process that can generate reactive carbonyls including formaldehyde, acetaldehyde, and acrolein, alongside high concentrations of ultrafine particles capable of penetrating distal airways.[5] Although often perceived as less harmful than combustible tobacco, accumulating clinical and epidemiologic evidence indicates that exclusive e-cigarette use is associated with measurable pulmonary effects, including reductions in airflow indices and increased respiratory symptom burden in otherwise young populations.[6,7]

Despite growing evidence, direct three-group comparative studies simultaneously assessing tobacco smokers, exclusive e-cigarette users, and non-smokers using standardised spirometry within a single cohort remain limited. Young adulthood represents a critical physiological window: impairment in maximally attained lung function during this period may shift the life-course respiratory trajectory downward and increase later obstructive lung disease risk.[8,9] Against this background, the present study was designed to compare pulmonary function parameters — FVC, FEV_1,

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FEV_1/FVC, PEFR, and $FEF_{25–75}$ — alongside spirometric pattern distribution and respiratory symptom burden across non-smokers, tobacco smokers, and exclusive e-cigarette users in young adults, using ATS/ERS 2019 standardised spirometry.

2. Materials and Methods

2.1. Study design and setting

This was an analytical, observational cross-sectional study conducted in the Department of Physiology. The study was designed to compare spirometric parameters and respiratory symptom burden across three mutually exclusive nicotine exposure groups in young adults. Ethical approval was obtained from the Institutional Ethics Committee prior to data collection, and written informed consent was secured from every participant before any measurement was performed. All procedures adhered to the Declaration of Helsinki guidelines for research involving human subjects.

2.2. Study population and sampling

A total of 375 individuals were screened for eligibility, of whom 75 were excluded on predefined grounds: failure to meet inclusion criteria (n = 30), acute respiratory illness or spirometry contraindication (n = 25), refusal to provide consent (n = 10), and incomplete questionnaire or unacceptable spirometry manoeuvres (n = 10). The remaining 300 participants were enrolled through purposive convenience sampling and allocated equally into three groups: non-smokers (n = 100), tobacco smokers (n = 100), and exclusive e-cigarette users (n = 100). Equal group allocation was adopted to optimise statistical comparability and to ensure adequate power for one-way between-group analysis across multiple spirometric outcomes.

2.3. Inclusion and exclusion criteria

Non-smokers were enrolled if they had no current or prior history of tobacco or e-cigarette use and no acute respiratory illness at the time of testing. Tobacco smokers were enrolled if they were current combustible cigarette users with a documented exposure history including duration, daily cigarette count, and pack-years. E-cigarette users were enrolled if they were exclusive current vapers with a recorded usage history including duration and frequency of use, and no concurrent tobacco use. Device wattage, coil resistance, e-liquid nicotine formulation/concentration (including freebase versus nicotine salts), puff topography, and flavour-specific constituent data were not systematically captured in the study instrument. Participants were excluded if they had a physician-diagnosed chronic respiratory condition (asthma, COPD, bronchiectasis, or interstitial lung disease), uncontrolled cardiovascular disease, recent thoracoabdominal surgery, pregnancy, or an inability to perform reproducible forced expiratory manoeuvres.

2.4. Sample size justification

The planned total sample of N = 300 was determined based on one-way ANOVA power calculations for three-group spirometric comparisons. Under conservative assumptions (Cohen's f = 0.20–0.25; α = 0.05; power = 0.80; k = 3), the minimum required total sample ranged from 158 to 244 participants. The target of 300 (100 per group) was selected to provide adequate power even for subtler small-airway indices ($FEF_{25–75}$), to accommodate covariate-adjusted secondary analyses, and to buffer against loss from unacceptable spirometry manoeuvres.

2.5. Data collection instruments

A structured case record form was used to collect demographic data (age, sex), anthropometric parameters (height by stadiometer, weight by calibrated scale, BMI), baseline vital signs (pulse rate, systolic and diastolic blood pressure by validated sphygmomanometer, respiratory rate), nicotine exposure history, and a standardised five-item respiratory symptom checklist (chronic cough, sputum, wheeze, chest tightness, exertional dyspnoea). Environmental and lifestyle confounders assessed included biomass fuel exposure, indoor cooking smoke, occupational dust or chemical exposure, alcohol use, and physical activity level, all recorded as binary categorical variables. Specific vaping flavours and flavour-related toxicants of interest (for example, cinnamaldehyde- or diacetyl-containing products) were not separately recorded.

2.6. Spirometry protocol

Spirometry was performed using a calibrated spirometer following American Thoracic Society/European Respiratory Society (ATS/ERS) standardisation principles and the 2019 technical update.[10,11] Participants were seated upright, instructed to wear loose clothing, and asked to refrain from smoking or vaping for a minimum of four hours before testing. A disposable mouthpiece and nose clip were used for each participant. Participants performed a maximum inhalation to total lung capacity followed by a rapid, forceful, sustained exhalation until complete emptying was confirmed. At least three acceptable manoeuvres were obtained, with reproducibility assessed using best-of-three selection. Primary spirometric indices recorded were FVC (L), FEV_1 (L), FEV_1/FVC (%), PEFR (L/min), and $FEF_{25–75}$ (L/s), together with FEV_1 %predicted and FVC %predicted. Spirometric patterns were classified as normal, obstructive (FEV_1/FVC < 0.70 or below the lower limit of normal [LLN]), restrictive (FVC < LLN with preserved FEV_1/FVC), or mixed (both criteria fulfilled), in accordance with ATS/ERS guidance and GLI-based reference interpretation.[11,12] The analytic dataset retained the final spirometric category assigned to each participant; however, it did not preserve a separate case-level flag indicating whether obstruction was triggered by the fixed-ratio threshold or the LLN criterion.

2.7. Statistical analysis

All analyses were performed using SPSS (version 25.0; IBM Corp., Armonk, NY). Continuous variables were expressed as mean ± standard deviation (SD) with 95% confidence intervals (CI). Categorical variables were presented as frequencies and percentages. Normality was assessed by the Shapiro–Wilk test, and variance homogeneity across groups was evaluated by Levene's test. Between-group differences for continuous variables were analysed by one-way ANOVA; where Levene's test indicated unequal variances ($p < 0.05$), the Welch-corrected F statistic was applied.[13] Pairwise post-hoc comparisons were conducted

with Bonferroni adjustment to control the family-wise type I error rate. Categorical variables were compared by Pearson chi-square test, with Fisher's exact test reserved for cells with expected counts below five.[14] Effect sizes were quantified using eta-squared (η^2) for ANOVA outcomes and Cohen's d for pairwise contrasts; Cramér's V was reported for categorical associations. A two-tailed p-value < 0.05 was considered statistically significant throughout.

CONSORT-style participant flow diagram (Figure 1)

3. Results

3.1. Participant characteristics, demographics, and exposure profile

A total of 375 individuals were screened, of whom 75 were excluded owing to acute illness, spirometry contraindication, incomplete questionnaires, or declining to participate. The final analytic cohort comprised 300 young adults allocated equally across three groups: non-smokers (n = 100), tobacco smokers (n = 100), and e-cigarette users (n = 100). Demographic and anthropometric parameters were comparable across all three groups (Table 1). Mean age was 29.83 ± 6.21 years in non-smokers, 31.02 ± 6.41 years in tobacco smokers, and 29.94 ± 7.00 years in e-cigarette users, with no significant between-group difference (F = 1.005; p = 0.366). The sample was predominantly male (non-smokers 95%, tobacco smokers 94%, e-cigarette users 96%; χ^2 = 0.42; p = 0.810), and height, weight, and BMI did not differ significantly between groups (all ANOVA p > 0.05; $\eta^2 \leq 0.006$), confirming that body habitus is an unlikely confounder of the observed spirometric differences.

Tobacco smokers had a mean smoking duration of 14.64 ± 6.30 years, a daily consumption of 10.18 ± 5.45 cigarettes, and a cumulative pack-year exposure of 7.27 ± 5.05 pack-years. Among exclusive e-cigarette users, duration and self-reported frequency of use were recorded for eligibility and exposure classification; however, device wattage, nicotine formulation/concentration, and flavour-specific constituent data were not systematically retained in the final analytic dataset.

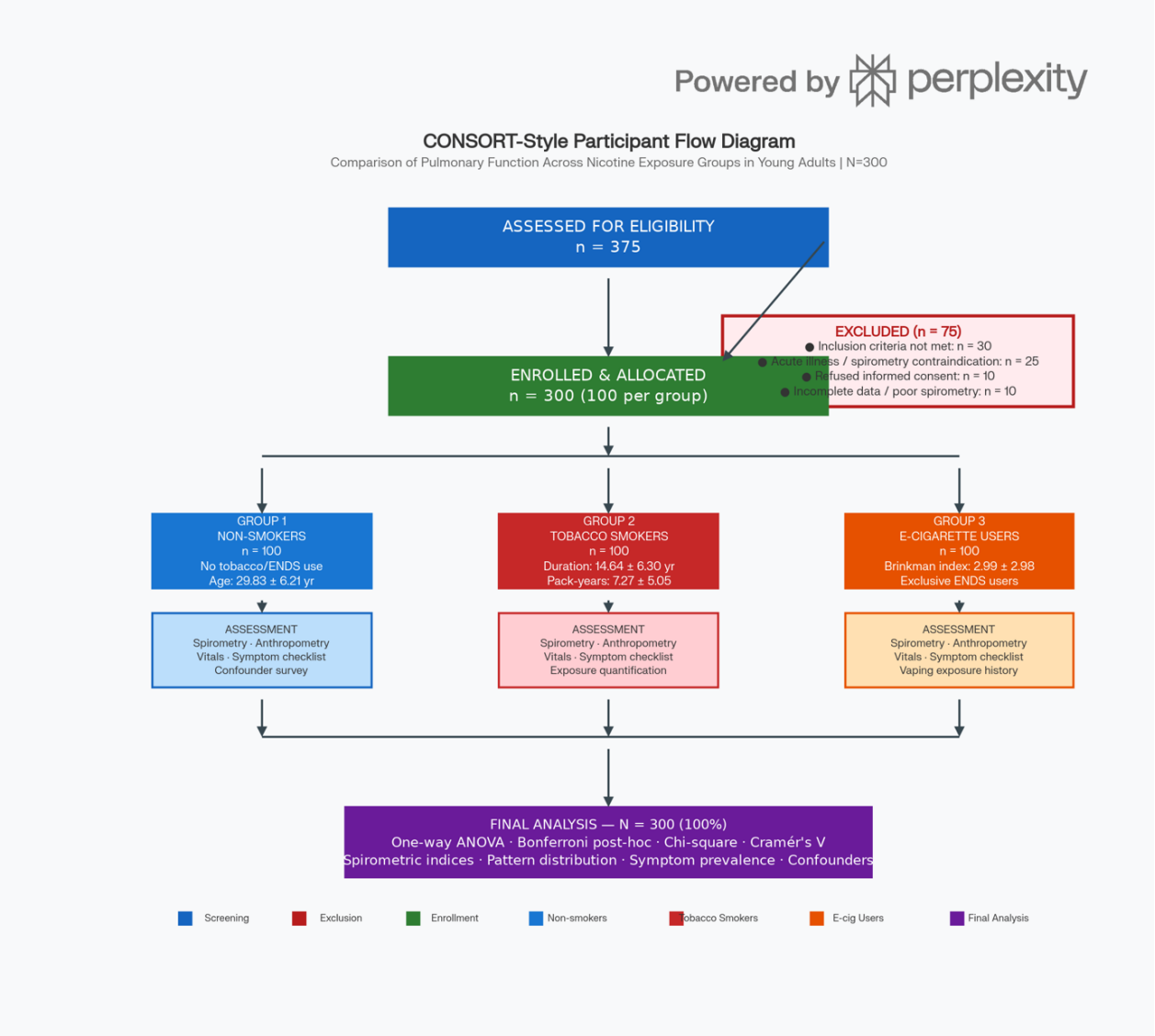


Figure 1. *CONSORT-style flow diagram of participant recruitment, screening, exclusion, allocation, and final analysis. All 300 enrolled participants completed the full assessment protocol and were included in the final analytic dataset.*

Table 1. *Participant Characteristics: Demographics, Anthropometry, Vital Signs, and Exposure Profile (N = 300)*

Variable	Non-smokers (n = 100)	Tobacco Smokers (n = 100)	E-cigarette Users (n = 100)	Statistic	p-value
Demographic Parameters					
Age (years), mean ± SD	29.83 ± 6.21	31.02 ± 6.41	29.94 ± 7.00	F = 1.005	0.366
Male sex, n (%)	95 (95.0%)	94 (94.0%)	96 (96.0%)	χ^2 = 0.42	0.810
Female sex, n (%)	5 (5.0%)	6 (6.0%)	4 (4.0%)	Cramér's V = 0.037	
Anthropometric Parameters					
Height (cm), mean ± SD	168.05 ± 6.58	169.34 ± 6.89	168.82 ± 6.56	F = 0.947	0.389
Weight (kg), mean ± SD	67.99 ± 9.94	68.22 ± 9.90	67.60 ± 9.58	F = 0.025	0.975
BMI (kg/m^2), mean ± SD	24.09 ± 3.33	23.82 ± 3.28	23.79 ± 3.57	F = 0.241	0.786
Baseline Vital Parameters					
Pulse rate (bpm), mean ± SD	76.93 ± 10.46	78.19 ± 11.14	76.09 ± 10.80	F = 0.957	0.385
Systolic BP (mmHg), mean ± SD	118.07 ± 10.80	123.86 ± 9.85	122.59 ± 10.10	Welch F	< 0.001
Diastolic BP (mmHg), mean ± SD	77.00 ± 7.70	78.95 ± 7.70	78.12 ± 7.38	F = 2.098	0.124
Respiratory rate (/min), mean ± SD	16.08 ± 1.74	16.99 ± 1.96	16.47 ± 1.84	Welch F	0.003
Nicotine Exposure Profile					
Duration of tobacco use (years)	—	14.64 ± 6.30	—	—	—
Cigarettes per day, n	—	10.18 ± 5.45	—	—	—
Pack-years (mean ± SD)	—	7.27 ± 5.05	—	—	—
E-cigarette device/liquid data	—	—	Not recorded	—	—

Values are mean ± SD or n (%). BMI = body mass index; BP = blood pressure. Bold = significant between-group difference (p < 0.05) by one-way ANOVA or Welch correction. Pack-years = cigarettes/day ÷ 20 × years smoked. Detailed e-cigarette device/liquid variables were not systematically recorded.

Table 2. *Pulmonary function parameters and percent-predicted values across exposure Groups (N = 300)*

Parameter	Non-smokers (n = 100) Mean ± SD [95% CI]	Tobacco Smokers (n = 100) Mean ± SD [95% CI]	E-cigarette Users (n = 100) Mean ± SD [95% CI]	Δ Smokers vs NS	Δ E-cig vs NS
FVC (L)	3.688 ± 0.831 [3.523–3.853]	3.559 ± 0.838 [3.393–3.725]	3.546 ± 0.914 [3.365–3.727]	−3.5%	−3.9%
FEV_1 (L)	3.085 ± 0.684 [2.949–3.221]	2.647 ± 0.712 [2.506–2.788]	2.822 ± 0.762 [2.671–2.973]	−14.2%	−8.5%
FEV_1/FVC (%)	83.78 ± 3.48 [83.09–84.47]	74.22 ± 7.86 [72.66–75.78]	79.57 ± 5.70 [78.44–80.70]	−9.56 pp	−4.21 pp
PEFR (L/min)	418.05 ± 44.11 [409.30–426.80]	408.30 ± 49.12 [398.55–418.05]	418.50 ± 55.21 [407.55–429.45]	−2.3%	+0.1%
FEF_{25-75} (L/s)	3.106 ± 0.716 [2.964–3.248]	2.257 ± 0.695 [2.119–2.395]	2.607 ± 0.652 [2.478–2.736]	−27.3%	−16.1%
FEV_1 %pred	99.99 ± 21.34 [95.76–104.22]	85.08 ± 22.41 [80.63–89.53]	90.56 ± 22.80 [86.04–95.08]	−14.9 pp	−9.4 pp
FVC %pred	99.99 ± 21.47 [95.73–104.25]	95.71 ± 21.82 [91.38–100.04]	95.18 ± 22.78 [90.66–99.70]	−4.3 pp	−4.8 pp

Values are mean ± SD [95% CI]. NS = non-smokers; pp = percentage points; Δ = absolute change from non-smoker group. Bold = clinically and statistically significant reductions (p < 0.05 in ANOVA). Spirometry performed per ATS/ERS 2019 standards. Lower FEV_1/FVC values indicate greater airflow limitation; formal obstruction classification in this study used FEV_1/FVC < 0.70 or < LLN rather than a fixed 80% threshold. FEF_{25-75} reflects small-airway function.

3.2. Pulmonary function parameters and percent-predicted values

Spirometric assessment revealed a consistent and clinically meaningful three-tier gradient in airflow indices across the three groups (Table 2). Non-smokers maintained fully preserved pulmonary function, with mean FEV_1 of 3.085 ± 0.684 L (95% CI: 2.949–3.221), FEV_1/FVC of 83.78 ± 3.48% (95% CI: 83.09–84.47%), and FEF_{25-75} of 3.106 ± 0.716 L/s (95% CI: 2.964–3.248), consistent with normal large- and small-airway function. Tobacco smokers showed the greatest impairment: FEV_1 was reduced to 2.647 ± 0.712 L (−14.2% vs non-smokers; Δ = −0.438 L), FEV_1/FVC to 74.22 ± 7.86% (−9.56 percentage points), and FEF_{25-75} to 2.257 ± 0.695 L/s (−27.3%; Δ = −0.849 L/s). E-cigarette users demonstrated intermediate impairment: FEV_1 of 2.822 ± 0.762 L (−8.5%; Δ = −0.263 L), FEV_1/FVC of 79.57 ± 5.70% (−4.21 pp), and FEF_{25-75} of 2.607 ± 0.652 L/s (−16.1%; Δ = −0.499 L/s). In contrast, FVC (p = 0.437; η^2 = 0.006) and PEFR (p = 0.262; η^2 = 0.009) did not differ significantly across groups, confirming a predominantly obstructive and small-airway functional pattern rather than restrictive volume loss.

Percent-predicted analysis reinforced this gradient: non-smokers had FEV_1 %pred of 99.99 ± 21.34%, e-cigarette users 90.56 ± 22.80%, and tobacco smokers 85.08 ± 22.41%, representing decrements of approximately 9.4 and 14.9 percentage points from the non-smoker reference, respectively. FVC %pred was comparatively preserved in both exposed groups (tobacco smokers: 95.71 ± 21.82%; e-cigarette users: 95.18 ± 22.78%), further confirming that volume loss is not the dominant abnormality and that flow-based indices are the primary indices of impairment in this cohort.

3.3. Between-group statistical analysis

One-way ANOVA and Welch-corrected tests confirmed significant between-group differences for FEV_1, FEV_1/FVC,

and $FEF_{25–75}$, while FVC and PEFR did not reach significance (Table 3). The largest effect was observed for FEV_1/FVC (F = 64.852; Welch F = 69.930; p < 0.001; η^2 = 0.304), indicating that exposure group accounts for approximately 30% of total variance in the airflow ratio — a large effect by conventional benchmarks ($\eta^2 \geq 0.14$). $FEF_{25–75}$ similarly yielded a large effect (F = 38.439; p < 0.001; η^2 = 0.206), confirming robust small-airway signal across groups. FEV_1 showed a moderate effect (F = 9.406; p < 0.001; η^2 = 0.060), while FVC (η^2 = 0.006) and PEFR (η^2 = 0.009) were negligible and non-significant, reinforcing that the primary functional impairment is obstructive and flow-based, not volumetric.

Bonferroni-adjusted post-hoc testing confirmed a consistent and statistically significant three-tier gradient for both FEV_1/FVC and $FEF_{25–75}$ across all three pairwise contrasts (Table 3). The non-smoker versus tobacco smoker contrast yielded the largest effect for FEV_1/FVC (mean difference = 9.56 pp; 95% CI: 7.86–11.26; p[Bonf] < 0.001; Cohen's d = 1.574 — very large effect) and for $FEF_{25–75}$ (Δ = 0.849 L/s; d = 1.203). Critically, the e-cigarette users versus tobacco smoker contrast was also independently significant for FEV_1/FVC (Δ = −5.35 pp; p[Bonf] < 0.001; d = −0.780) and $FEF_{25–75}$ (Δ = −0.350 L/s; p[Bonf] = 0.001; d = −0.519), demonstrating that the two nicotine-exposed groups are statistically distinguishable from each other and not merely from the control group. FVC and PEFR showed no significant pairwise differences after Bonferroni correction.

3.4. Spirometric pattern distribution

The distribution of spirometric patterns differed markedly and significantly across groups (χ^2 = 59.40, df = 6, p < 0.001; Cramér's V = 0.315) (Table 4). The analytic dataset retained final spirometric categories but not the case-level breakdown of whether the obstructive label was triggered by the fixed-ratio criterion, the LLN criterion, or both; criterion-specific frequencies therefore cannot be reported for the obstructive subgroup. Among non-smokers, 94% demonstrated entirely normal spirometry, with obstructive and restrictive patterns occurring in only 4% and 2% respectively, and no mixed defects. Tobacco smokers showed the greatest burden of abnormal spirometry: only 46% were classified as normal, while 43% had obstructive, 7% restrictive, and 4% mixed patterns, representing a more than ten-fold increase in obstructive prevalence compared with non-smokers. E-cigarette users occupied an intermediate position: 70% normal, 24% obstructive, and 6% restrictive, with no mixed patterns, with a six-fold higher obstructive prevalence compared with non-smokers — a pattern that precisely mirrors the quantitative spirometric gradient.

Table 3. *Between-group ANOVA and bonferroni Post-Hoc pairwise comparisons (N = 300)*

Parameter	ANOVA F	Welch F	p-value	η^2	Comparison	Mean Diff.	95% CI	p (Bonf.)	Cohen's d
FVC (L)	0.830	0.852	0.437	0.006	NS vs Smokers	0.129	−0.104 to 0.362	0.825	0.155
					NS vs E-cig	0.142	−0.102 to 0.385	0.756	0.162
					Smokers vs E-cig	−0.013	−0.257 to 0.232	1.000	−0.015
FEV_1 (L)	9.406	10.007	< 0.001	0.060	NS vs Smokers	0.438	0.244 to 0.633	< 0.001	0.628
					NS vs E-cig	0.263	0.062 to 0.465	0.032	0.364
					Smokers vs E-cig	−0.175	−0.381 to 0.030	0.284	−0.238
FEV_1/FVC (%)	64.852	69.930	< 0.001	0.304	NS vs Smokers	9.563	7.864 to 11.262	< 0.001	1.574
					NS vs E-cig	4.214	2.896 to 5.532	< 0.001	0.893
					Smokers vs E-cig	−5.349	−7.264 to −3.434	< 0.001	−0.780
PEFR (L/min)	1.346	1.366	0.262	0.009	NS vs Smokers	9.750	−3.269 to 22.769	0.424	0.209
					NS vs E-cig	−0.450	−14.389 to 13.489	1.000	−0.009
					Smokers vs E-cig	−10.200	−24.773 to 4.373	0.507	−0.195
$FEF_{25–75}$ (L/s)	38.439	36.367	< 0.001	0.206	NS vs Smokers	0.849	0.652 to 1.046	< 0.001	1.203
					NS vs E-cig	0.499	0.308 to 0.690	< 0.001	0.729
					Smokers vs E-cig	−0.350	−0.538 to −0.162	0.001	−0.519

Bold = statistically significant (p < 0.05). NS = non-smokers. η^2 interpretation: < 0.01 negligible, 0.01–0.06 small, 0.06–0.14 moderate, ≥ 0.14 large. Cohen's d: 0.2 small, 0.5 medium, 0.8 large. Welch F used where Levene's test p < 0.05 (FEV_1/FVC). Mean differences represent first group minus second group listed; positive = first group higher.

Table 4. *Spirometric pattern distribution across exposure groups (N = 300)*

Spirometric Pattern	Non-smokers (n = 100) n (%)	Tobacco Smokers (n = 100) n (%)	E-cigarette Users (n = 100) n (%)	χ^2	p-value	Cramér's V
Normal	94 (94.0%)	46 (46.0%)	70 (70.0%)	59.40	< 0.001	0.315
Obstructive	4 (4.0%)	43 (43.0%)	24 (24.0%)			
Restrictive	2 (2.0%)	7 (7.0%)	6 (6.0%)			
Mixed	0 (0.0%)	4 (4.0%)	0 (0.0%)			
Any abnormal pattern	6 (6.0%)	54 (54.0%)	30 (30.0%)			

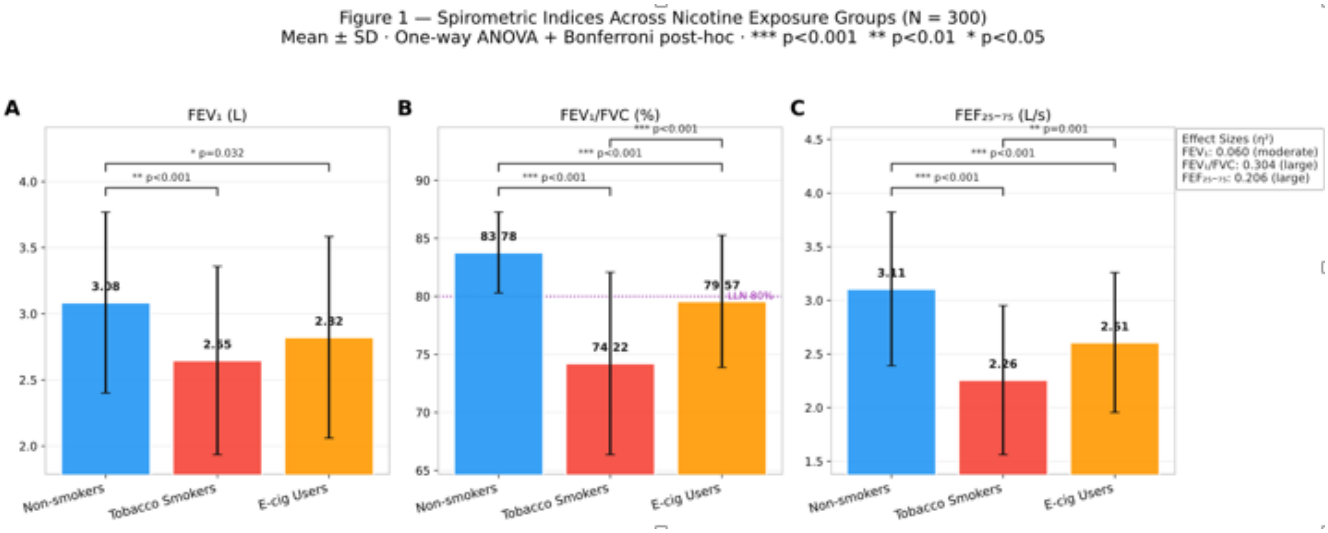


Figure 2. *Comparison of key spirometric indices across nicotine exposure groups in young adults*

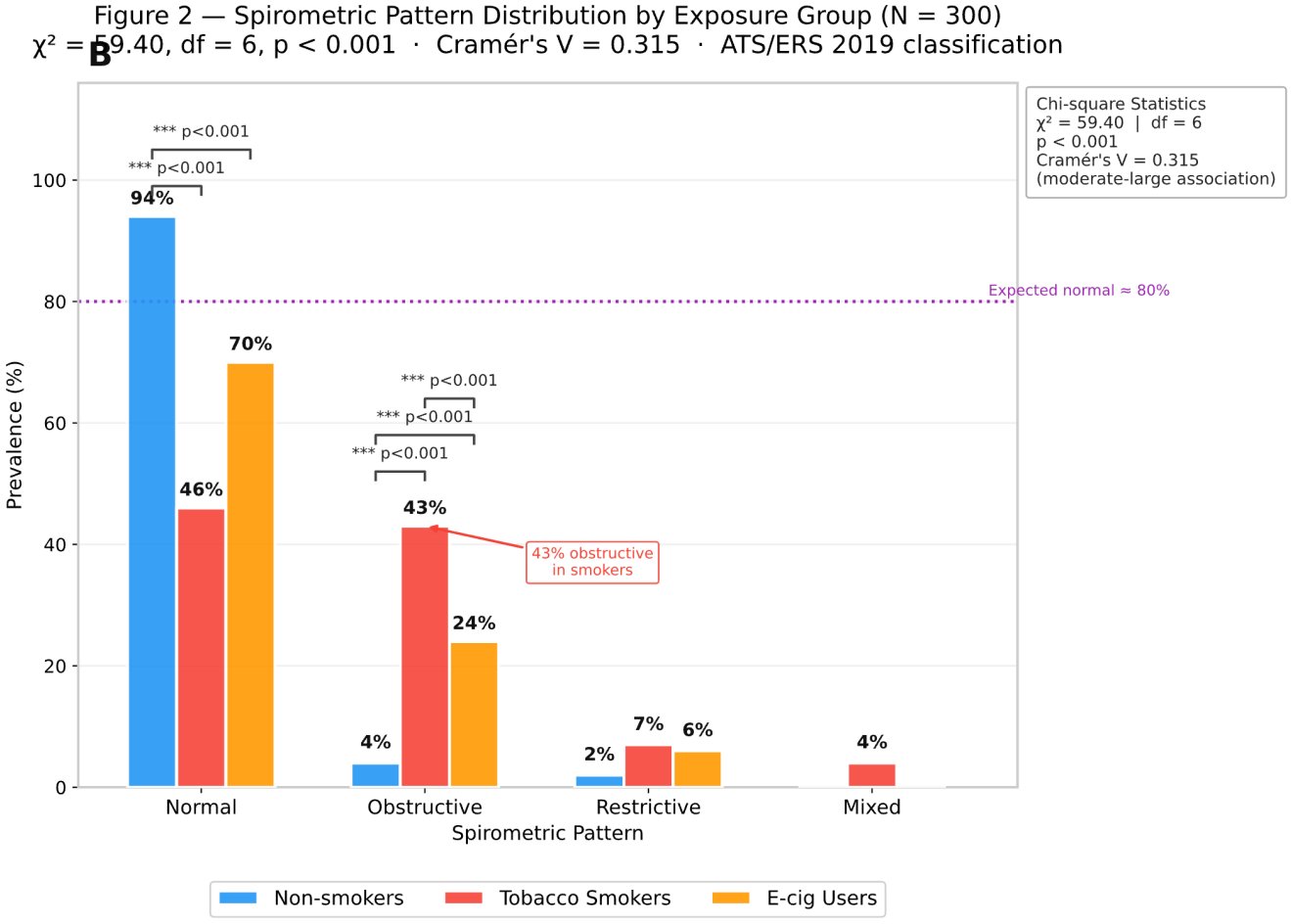


Figure 3. *Distribution of spirometric patterns by nicotine exposure group in young adults*

Spirometric patterns classified per ATS/ERS 2019 criteria. Obstruction: $FEV_1/FVC < 0.70$ or $< LLN$; restriction: $FVC < LLN$ with normal FEV_1/FVC; mixed: both criteria met. $\chi^2 = 59.40$, df = 6; Cramér's V = 0.315 indicates a moderate-to-large association between exposure group and spirometric classification. Abnormal pattern prevalence: tobacco smokers 54% > e-cigarette users 30% > non-smokers 6%. Because the analytic dataset stored final spirometric categories rather than criterion-specific trigger flags, the relative contribution of fixed-ratio versus LLN classification within the obstructive category could not be disaggregated retrospectively.

Figure 2 illustrating mean (± standard deviation) values of forced expiratory volume in one second (FEV_1, L), FEV_1/forced vital capacity ratio (FEV_1/FVC, %), and forced expiratory flow at 25–75% of FVC ($FEF_{25–75}$,

L/s) across three mutually exclusive nicotine exposure groups: non-smokers (blue; n = 100), tobacco smokers (red; n = 100), and exclusive e-cigarette users (orange; n = 100); total N = 300. Between-group differences were evaluated by one-way analysis of variance (ANOVA) with Welch correction for variance heterogeneity; pairwise comparisons were adjusted using the Bonferroni method. Tobacco smokers demonstrated the greatest reductions in FEV_1 (−14.2%), FEV_1/FVC (−9.56 percentage points), and $FEF_{25–75}$ (−27.3%) relative to non-smokers, while e-cigarette users showed intermediate impairment (FEV_1: −8.5%; FEV_1/FVC: −4.21 pp; $FEF_{25–75}$: −16.1%). The FEV_1/FVC and $FEF_{25–75}$ contrasts reached statistical significance across all three pairwise comparisons. Effect sizes were large for FEV_1/FVC (η^2 = 0.304) and $FEF_{25–75}$ (η^2 = 0.206). Because LLN varies according to age, sex, height, and reference equation, no single fixed LLN line should be inferred from this illustration. Spirometry was performed in accordance with American Thoracic Society/European Respiratory Society (ATS/ERS) 2019 standards. Statistical significance: *p < 0.05; **p < 0.01; ***p < 0.001.

Figure 3 depicting the percentage distribution of spirometric patterns — normal, obstructive, restrictive, and mixed — across non-smokers (blue; n = 100), tobacco smokers (red; n = 100), and exclusive e-cigarette users (orange; n = 100); total N = 300. Spirometric patterns were classified in accordance with ATS/ERS 2019 criteria: obstruction defined as FEV_1/FVC < 0.70 or below the lower limit of normal (LLN); restriction defined as FVC < LLN with a preserved FEV_1/FVC ratio; mixed pattern defined as both criteria fulfilled simultaneously. The distribution of spirometric patterns differed significantly across groups (Pearson χ^2 = 59.40, df = 6, p < 0.001; Cramér's V = 0.315), indicating a moderate-to-large group–pattern association. Normal spirometry was most prevalent in non-smokers (94%), intermediate in e-cigarette users (70%), and least frequent in tobacco smokers (46%). Conversely, obstructive patterns were most common in tobacco smokers (43%), followed by e-cigarette users (24%), and least frequent in non-smokers (4%). The final analytic dataset retained overall spirometric pattern classification but not the criterion-specific breakdown of fixed-ratio versus LLN triggers within obstructive cases; accordingly, only the aggregate obstructive prevalence is displayed. Percentage labels are displayed above each bar. Chi-square statistics are displayed in the upper right panel inset.

3.5. Respiratory symptoms across groups

Respiratory symptom prevalence was assessed for cough, sputum production, wheeze, chest tightness, and exertional dyspnoea (Table 5). Wheeze showed the strongest trend toward differential prevalence — 70% in tobacco smokers, 62% in e-cigarette users, and 55% in non-smokers (χ^2 = 4.800; p = 0.091; Cramér's V = 0.126) — but did not reach Bonferroni-corrected significance. Cough, sputum, chest tightness, and dyspnoea were similarly distributed across all three groups (all χ^2 p $\geq$ 0.340), a finding consistent with the well-established observation that spirometrically detectable airflow limitation, particularly early small-airway dysfunction, precedes overt symptom differentiation in young adults. The absence of significant symptom differences despite clear spirometric gradients reinforces the clinical value of objective spirometric screening in this age group.

Table 5. *Respiratory symptom prevalence by exposure group (N = 300)*

Symptom	Non-smokers (n = 100) n (%)	Tobacco Smokers (n = 100) n (%)	E-cigarette Users (n = 100) n (%)	χ^2	p-value	Cramér's V
Chronic cough	56 (56.0%)	47 (47.0%)	47 (47.0%)	2.160	0.340	0.085
Sputum production	49 (49.0%)	52 (52.0%)	52 (52.0%)	0.240	0.887	0.028
Wheeze	55 (55.0%)	70 (70.0%)	62 (62.0%)	4.800	0.091	0.126
Chest tightness	74 (74.0%)	78 (78.0%)	74 (74.0%)	0.570	0.751	0.044
Exertional dyspnoea	50 (50.0%)	54 (54.0%)	49 (49.0%)	0.560	0.756	0.043

Chi-square test used for all comparisons. No symptom reached Bonferroni-corrected significance threshold (α = 0.0167 for three primary comparisons). Cramér's V < 0.13 for all symptoms indicates weak group–symptom association. The absence of significant symptom differences despite significant spirometric gradients is consistent with subclinical early-phase airway dysfunction preceding symptom onset.

Table 6. *Environmental and lifestyle confounders by exposure group (N = 300)*

Confounder	Non-smokers (n = 100) n (%)	Tobacco Smokers (n = 100) n (%)	E-cigarette Users (n = 100) n (%)	χ^2	p-value	Cramér's V
Biomass fuel exposure	56 (56.0%)	51 (51.0%)	52 (52.0%)	0.560	0.755	0.043
Indoor cooking smoke	51 (51.0%)	50 (50.0%)	57 (57.0%)	1.150	0.563	0.062
Occupational dust/chemical exposure	47 (47.0%)	50 (50.0%)	47 (47.0%)	0.240	0.887	0.028
Alcohol use (any)	45 (45.0%)	49 (49.0%)	50 (50.0%)	0.560	0.755	0.043
Passive smoke exposure	0 (0.0%)	0 (0.0%)	0 (0.0%)	—	—	—
Physical activity — Sedentary	57 (57.0%)	48 (48.0%)	49 (49.0%)	1.950	0.378	0.081
Physical activity — Moderate/Active	43 (43.0%)	52 (52.0%)	51 (51.0%)	1.950	0.378	0.081

All comparisons by chi-square test. Cramér's V < 0.10 for all variables indicates negligible association between group membership and any confounder, confirming balanced distribution and supporting internal validity of the study.

3.6. Environmental and lifestyle confounders

All measured environmental and lifestyle confounders were broadly and comparably distributed across the three groups, with no variable reaching statistical significance (Table 6). Biomass fuel exposure (56% vs 51% vs 52%; p = 0.755), indoor cooking smoke (51% vs 50% vs 57%; p = 0.563), occupational dust or chemical exposure (47% vs 50% vs 47%; p = 0.887), alcohol use (45% vs 49% vs 50%; p = 0.755), and physical activity level (sedentary: 57% vs 48% vs 49%; p = 0.378) all showed Cramér's V $\leq$ 0.081, indicating negligible association with group membership. This homogeneous confounder distribution substantially strengthens the internal validity of the between-group spirometric comparisons, supporting the inference that the observed pulmonary function gradient is attributable to tobacco or e-cigarette exposure rather than to measured environmental or behavioural imbalance.

4. Discussion

This comparative cross-sectional study demonstrated a statistically significant, dose-ordered three-tier gradient in pulmonary function across non-smokers, e-cigarette users, and tobacco smokers in young adults, with tobacco smokers exhibiting the greatest impairment, e-cigarette users showing intermediate dysfunction, and non-smokers maintaining fully preserved spirometry. The primary abnormality was flow-based and obstructive, centred on FEV_1/FVC and FEF_{25-75}, while FVC and PEFR remained non-significantly different between groups — a pattern consistent with peripheral small-airway injury preceding global ventilatory restriction.

The reduction in FEV_1/FVC to 74.22 $\pm$ 7.86% among tobacco smokers (η^2 = 0.304; Cohen's d = 1.574) is directionally consistent with longitudinal and observational evidence linking cumulative smoking exposure with accelerated decline in expiratory airflow and adverse lung-function trajectories. Oelsner et al. showed accelerated FEV_1 decline in former smokers and current smokers, including those with low cumulative exposure, while Chowdhury et al. reported an inverse relationship between pack-years of smoking and spirometric indices including FEV_1, FEV_1/FVC, and PEFR in asymptomatic smokers.[3,15]

The intermediate spirometric profile documented in e-cigarette users — FEV_1/FVC of 79.57 $\pm$ 5.70% (Δ = −4.21 pp; d = 0.893) and FEF_{25-75} of 2.607 $\pm$ 0.652 L/s (Δ = −16.1%) — aligns with accumulating evidence that electronic nicotine delivery systems are not physiologically inert. Clinical studies in healthy adults have reported acute and subacute changes in lung function and exhaled nitric oxide after e-cigarette exposure, while contemporary reviews conclude that respiratory effects are biologically plausible through airway irritation, oxidative stress, inflammation, and small-airway toxicant deposition.[6,7]

The obstructive pattern prevalence of 43% in tobacco smokers versus 24% in e-cigarette users and 4% in non-smokers (χ^2 = 59.40; Cramér's V = 0.315) is consistent with a broader risk hierarchy in which combustible tobacco shows the greatest respiratory burden, e-cigarette exposure shows intermediate risk, and never-users have the lowest risk. Longitudinal PATH analyses in U.S. young adults and adults have linked exclusive e-cigarette use and other tobacco product use patterns to respiratory symptom development and broader adverse respiratory outcomes, with risk generally lower than that of combustible smoking and higher among multi-product users.[16,20–22] The absence of significant respiratory symptom differences despite clear spirometric gradients in the present cohort is compatible with the natural history of early small-airway disease, in which physiologic airflow impairment may precede overt clinical symptom differentiation.[9]

The homogeneous distribution of measured environmental confounders — biomass fuel exposure, indoor cooking smoke, occupational dust, alcohol use, and physical activity level (all p > 0.05; Cramér's V $\leq$ 0.081) — strengthens the internal comparability of the three groups, but does not eliminate the possibility of residual confounding.

5. Study Limitations

This study has several limitations relevant to interpretation. First, its cross-sectional design precludes causal inference. Second, vaping exposure was characterised only by use duration and frequency; device wattage, coil resistance, puff topography, e-liquid nicotine concentration, and nicotine formulation (freebase versus nicotine salts) were not systematically captured. Third, specific flavour categories and flavouring toxicants of clinical interest, including cinnamaldehyde- or diacetyl-containing products, were not recorded. Fourth, although spirometric obstruction was defined using FEV_1/FVC < 0.70 or < LLN, the analytic dataset retained only the final spirometric category and did not preserve a case-level flag identifying which criterion triggered classification for each obstructive participant. These limitations should be considered when interpreting the 24% obstructive prevalence in exclusive e-cigarette users and the observed reduction in FEF_{25-75} (5,17-19,23). Prospective studies with granular vaping phenotyping and criterion-level spirometric archiving are warranted.

6. Conclusion

This study shows that both tobacco smoking and e-cigarette use are associated with measurable spirometric impairment in young adults, following a consistent non-smoker > e-cigarette > tobacco smoker gradient. Tobacco smokers demonstrated the greatest FEV_1/FVC reduction and highest obstructive pattern prevalence (43%), while e-cigarette users exhibited significant intermediate impairment (24% obstructive). The selective reduction in FEF_{25-75} across both exposed groups is compatible with early peripheral airway dysfunction preceding clear symptom differentiation. Given the cross-sectional design and the absence of granular device/liquid exposure data, these findings should be interpreted cautiously but support consideration of spirometric screening in young nicotine-exposed individuals.

7. Source of Funding

None.

8. Conflict of Interest

None.

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